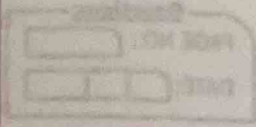


10/9



Emotions
PAGE NO.:
DATE:

MATHS

1- ~~2~~ 1) 3

2- 4) NO solution

3- 1) 10 units

4- 2) proportional

5- 2) $-\frac{1}{2}$

6- 1) $x^3 y^3$

7- $\frac{\alpha + \beta}{2} = \frac{\alpha \beta}{2}$

$$2\alpha + 2\beta = \alpha \beta$$

$$2(\alpha + \beta) = \alpha \beta$$

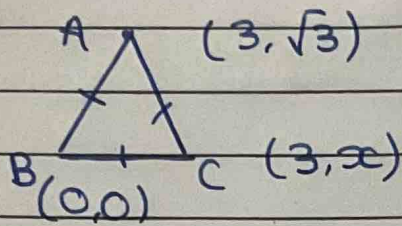
$$2 - \frac{b}{a} = \frac{c}{a}$$

$$2 \times \frac{2}{4} = \frac{K-4}{4}$$

$$4 = K - 4$$

$$\underline{\underline{K = 8}}$$

8.



$$\begin{aligned}
 AC &= AB \\
 \sqrt{(3-0)^2 + (\sqrt{3}-0)^2} &= \sqrt{(3-3)^2 + (x-\sqrt{3})^2} \\
 (0)^2 + (\sqrt{3})^2 &= 3^2 + (x-\sqrt{3})^2 \\
 x^2 + 3 + 2\sqrt{3}x &= 9 + 3 \\
 x^2 + 2\sqrt{3}x &= 6 \\
 x(x + 2\sqrt{3}) &= 6 \\
 x + 2\sqrt{3} &= 6/x
 \end{aligned}$$

$$AB^2 = AC^2$$

$$\begin{aligned}
 \Rightarrow (3-0)^2 + (\sqrt{3}-0)^2 &= (3-3)^2 + (x-\sqrt{3})^2 \\
 \Rightarrow (\sqrt{3})^2 &= (x)^2 \\
 \Rightarrow x &= \sqrt{3}
 \end{aligned}$$

9.

$$\begin{aligned}
 a_1 &\neq b_1 \\
 a_2 & \quad b_2
 \end{aligned}$$

$$\Rightarrow \frac{2p}{x} \neq \frac{3}{1}$$

$$\Rightarrow \underline{\underline{p \neq 3}}$$

10- In $\triangle ABC$, $DP \parallel CB$

By BPT, we get

$$\frac{AD}{DC} = \frac{AP}{PB} \quad - (1)$$

In $\triangle ABC$, $PE \parallel AC$

By BPT, we get

$$\frac{CE}{BE} = \frac{AP}{PB} \quad - (2)$$

On equating (1) & (2), we get

$$\frac{AD}{DC} = \frac{CE}{BE}$$

Hence, proved

11- $\tan \theta = \frac{1}{2}$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2} \Rightarrow \frac{P}{B} = \frac{1}{2}$$

$$P = 1$$

$$B = 2$$

$$H^2 = \sqrt{5}$$

$$\cos \theta = \frac{B}{H} = \frac{2}{\sqrt{5}}, \quad \sin \theta = \frac{P}{H} = \frac{1}{\sqrt{5}}$$

~~$$\sin \theta = \frac{P}{H} = \frac{1}{\sqrt{5}}$$~~

$$\frac{1}{2} \times \sin \theta \times \cos \theta$$

$$\Rightarrow \frac{1}{2} \times \frac{1}{\sqrt{3}} \times \frac{2}{\sqrt{3}}$$

$$\Rightarrow \frac{1}{3} = \text{RHS}$$

hence, proved

$$\begin{aligned} 12- \quad \frac{257}{5000} &= \frac{257}{5 \times 5 \times 2 \times 5 \times 2 \times 5 \times 2} \\ &= \underline{\underline{0.0514}} \end{aligned}$$

13- Let $\sqrt{3}$ be rational

$$\sqrt{3} = \frac{a}{b}$$

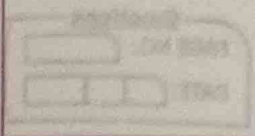
where 'a' and 'b' are co-primes & $b \neq 0$

Squaring both sides

$$3 = \frac{a^2}{b^2}$$

$$\frac{b^2}{3} = \frac{a^2}{3}$$

hence, 'a' is divisible by 3 - (1)



$$\text{Let } a = 3c$$

$$b^2 = \frac{(3c)^2}{3}$$

$$b^2 = 3c^2$$

$$c^2 = \frac{b^2}{3}$$

hence, 'b' is divis-
-ible by 3 - (2)

By (1) and (2) we get that both 'a' and 'b' are divisible by 3 which contradicts the fact that they are co-primes.

This contradiction has arisen because of our wrong assumption of $\sqrt{3}$ as rational.

$\therefore \sqrt{3}$ is irrational

hence, proved

$$14- \quad \alpha + \beta = -b/a$$

$$= 3/2$$

$$2\beta = c/a$$

$$= 7/2$$

$$\frac{1}{2\alpha-3} + \frac{1}{2\beta-3}$$

$$= \frac{(2\alpha-3) + (2\beta-3)}{(2\alpha-3)(2\beta-3)}$$

$$= \frac{\cancel{2\alpha} + \cancel{2\beta} - 3 - 3}{4\alpha\beta - 6\alpha - 6\beta + 9}$$

$$= \frac{2(\alpha + \beta) - 6}{4\alpha\beta - 6(\alpha + \beta) + 9}$$

$$= \frac{(2 \times 3/2) - 6}{(4 \times 7/2) - (6 \times 3/2) + 9}$$

$$= \frac{-3}{14 - 9 + 9}$$

$$= \frac{-3}{14}$$

15- $a_1 = b_1 = c_1$
 $a_2 = b_2 = c_2$

$$\frac{3}{5} = \frac{-a-1}{1-2a} = \frac{2b-1}{3b}$$

$$\frac{3}{5} = \frac{-a-1}{1-2a}$$

$$\Rightarrow 3-6a = -5a-5$$

$$\Rightarrow \underline{8=a}$$

$$\frac{3}{5} = \frac{2b-1}{3b}$$

$$\Rightarrow 9b = 10b-5$$

$$\Rightarrow \underline{b=5}$$

16- Given - $\triangle ABE \cong \triangle ACD$

To prove - $\triangle ADE \cong \triangle ABC$

Proof - $\triangle ABE \cong \triangle ACD$

$$\therefore AB = AC \quad (\text{CPCT}) - \textcircled{1}$$

$$AD = AE \quad (\text{CPCT}) - \textcircled{2}$$

In $\triangle ADE$ & $\triangle ABC$

$$\frac{AD}{AB} = \frac{AE}{AC} \quad (\textcircled{2} \div \textcircled{1})$$

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\triangle ADE \cong \triangle ABC$$

$$\angle A = \angle A \quad (\text{Common})$$

$$\therefore \triangle ABE \sim \triangle ABC \quad (\text{SAS})$$

Hence, proved

$$17. \quad \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\sec^2 \theta}{\sec^2 \theta - \cos^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

LHS

$$= \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$$

$$= \left(\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} \right) + \left(\frac{1}{\sin^2 \theta} \times \frac{-\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta \sin^2 \theta} \right)$$

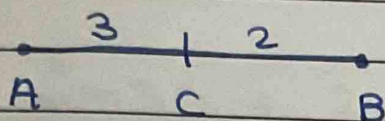
$$= \frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{1}{\sin^2 \theta} \times \frac{\cos^2 \theta \sin^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{1}{\sin^2 \theta - \cos^2 \theta} = \text{RHS}$$

Hence, proved

18



$$A \rightarrow (-2, 7)$$

$$B \rightarrow (x, y)$$

$$C \rightarrow (1, 1)$$

$$m_1 = 3$$

$$m_2 = 2$$

$$1, 1 = \left[\left(\frac{2x - 2 + 3 \times x}{3 + 2} \right), \left(\frac{2 \times 7 + 3 \times y}{3 + 2} \right) \right]$$

$$1, 1 = \left[\left(\frac{-4 + 3x}{5} \right), \left(\frac{14 + 3y}{5} \right) \right]$$

$$\Rightarrow 1 = \frac{-4 + 3x}{5}$$

$$\Rightarrow 5 = -4 + 3x$$

$$\Rightarrow 3x = 9$$

$$\Rightarrow \underline{x = 3}$$

$$C \rightarrow (3, -3)$$

$$1 = \frac{14 + 3y}{5}$$

$$\Rightarrow 3y = -9$$

$$\Rightarrow \underline{y = -3}$$

19. $\frac{x}{y} = \text{Fraction}$

$$\frac{2x}{y+1} = \frac{1}{1} \quad \& \quad \frac{x+4}{2y} = \frac{1}{2}$$



$$2x - y = 1 \quad \text{--- (1)}$$



$$\begin{aligned} 2x - 2y &= -8 \\ \Rightarrow 2(x - y) &= -8 \\ \Rightarrow x - y &= -4 \quad \text{--- (2)} \end{aligned}$$

$$\textcircled{1} - \textcircled{2}$$

$$\Rightarrow 2x - y = 1$$

$$- \quad x - y = -4$$

$$\begin{array}{r} \textcircled{-} \quad \textcircled{+} \quad \textcircled{+} \\ \hline \end{array}$$

$$\boxed{x = 5}$$

Put $x = 5$ in $\textcircled{2}$

$$x - y = -4$$

$$5 - y = -4$$

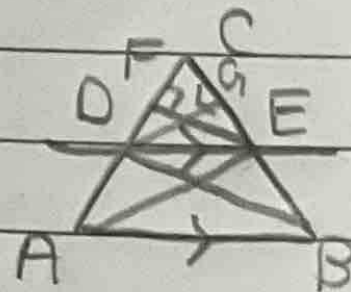
$$-y = -9$$

$$\boxed{y = 9}$$

20

If a line drawn parallel to any side of a triangle, then it divides the sides that it intersects in equal ratios.

Given - $AB \parallel DE$



To prove - $\frac{CD}{DA} = \frac{CE}{EB}$

Construction - Draw $EF \perp AC$
 Draw $DG \perp BC$
 Join AE
 Join BD

Proof - In $\triangle CDE$

$$Ar_1 = \frac{1}{2} (DE)(EF) \quad - (1)$$

$$Ar_2 = \frac{1}{2} (CE)(DG) \quad - (2)$$

In $\triangle ADE$

$$Ar = \frac{1}{2} (AD)(EF) \quad - (3)$$

In $\triangle BOE$

$$Ar = \frac{1}{2} (EB)(DG) \quad - (4)$$

$$(1) \div (3)$$

$$= \frac{Ar_1 (\triangle CDE)}{Ar (\triangle ADE)} = \frac{\cancel{\frac{1}{2}} (DE)(EF)}{\cancel{\frac{1}{2}} (AD)(EF)} = \frac{DC}{AD} \quad - (5)$$

$$(2) \div (4)$$

$$= \frac{Ar_2 (\triangle CDE)}{Ar (\triangle BOE)} = \frac{\cancel{\frac{1}{2}} (CE)(DG)}{\cancel{\frac{1}{2}} (EB)(DG)} = \frac{CE}{EB}$$

$$(1) = (2) \quad (\text{Area of same } \triangle)$$

$$\textcircled{3} = \textcircled{4}$$

(~~Two~~ Triangles between two parallel lines and having same base)

$$\therefore \frac{CD}{DA} = \frac{CE}{EB}$$

Hence, proved

20v

21-

$$\sin 30 + 2 \cos^2 45 + \tan^2 60$$

$$1/2 \cot 45 + \cos^2 30 + \tan^2 45$$

$$= \frac{1/2 + (1 \times 1/2) + 3}{1/2 \times 1 + 3/4 + 1}$$

$$= \frac{1/2 + 1 + 3}{1/2 + 3/4 + 1}$$

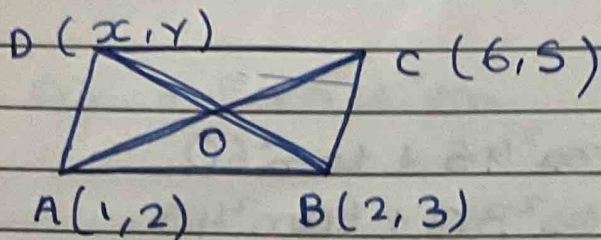
$$= \frac{9/2}{3/2 + 3/4}$$

$$= \frac{9/2}{9/4}$$

$$= \frac{9}{2} \times \frac{4}{9}$$

$$= \underline{\underline{2}}$$

22.



$$O \rightarrow \left(\frac{1+6}{2}, \frac{5+2}{2} \right)$$

$$O = \left(\frac{7}{2}, \frac{7}{2} \right)$$

$$\left(\frac{7}{2}, \frac{7}{2} \right) = \left(\frac{2+x}{2}, \frac{3+y}{2} \right)$$

$$\Rightarrow \frac{7}{2} = \frac{2+x}{2}$$

$$\Rightarrow \boxed{x = 5}$$

$$\Rightarrow \frac{7}{2} = \frac{3+y}{2}$$

$$\Rightarrow \boxed{y = 4}$$

$$\underline{D \rightarrow (5, 4)}$$

23- A $\rightarrow$ time to complete one round
 $= 1980/330$
 $= 6 \text{ min}$

B $\rightarrow = 10 \text{ min}$

C $\rightarrow = 9 \text{ min}$

Time of meeting = $\text{LCM}(6, 10, 9)$

$$\begin{array}{r|l} 2 & 6, 10, 9 \\ 3 & 3, 5, 9 \\ 3 & 1, 5, 3 \\ 5 & 1, 5, 3 \\ & 1, 1, 1 \end{array}$$

$2 \times 3 \times 3 \times 5$
 $= 90 \text{ mins later}$

24- Father's age = a
 Son's age = b

$2b + a = 70$ - (1)

$2a + b = 95$ - (2)

Multiply (2) by 2
 $\Rightarrow 4a + 2b = 190$ - (3)

~~(3) - (1)~~
 ~~$= 4a + 2b$~~

$$(3) - (1)$$

$$4a + 2b = 190$$

$$- a + 2b = 70$$

$$\begin{array}{r} \ominus \quad \ominus \quad \ominus \\ \hline \end{array}$$

$$3a = 120$$

$$a = 40 \text{ yrs}$$

$$\text{Put } a = 40 \text{ in } (1)$$

$$2b + a = 70$$

$$2b = 30$$

$$b = 15 \text{ yrs}$$

$$\text{Father's age} = 40 \text{ yrs}$$

$$\text{Son's age} = 15 \text{ yrs}$$

25-

$$A \rightarrow (1, -7)$$

$$B \rightarrow (6, 4)$$

$$C \rightarrow (x, 0)$$

$$m_1 = K$$

$$m_2 = 1$$

$$(x, 0) = \left(\frac{K \times 6 + 1 \times 1}{K+1}, \frac{K \times 4 + 1 \times -7}{K+1} \right)$$

$$0 = \frac{4K - 7}{K+1}$$

$$4K - 7 = 0$$

$$\Rightarrow 4K - 7 = 0$$

$$4K = 7$$

$$K = \frac{7}{4}$$

$$\text{Ratio} = \boxed{7:4}$$

$$26 - \csc \theta - \sin \theta = m$$

$$\sec \theta - \cos \theta = n$$

$$\frac{1}{\sin \theta} - \sin \theta = m$$

$$\Rightarrow \frac{1 - \sin^2 \theta}{\sin \theta} = m$$

$$\Rightarrow \frac{\cos^2 \theta}{\sin \theta} = m$$

$$\frac{1}{\cos \theta} - \cos \theta = n$$

$$\Rightarrow \frac{1 - \cos^2 \theta}{\cos \theta} = n$$

$$\Rightarrow \frac{\sin^2 \theta}{\cos \theta} = n$$

$$(\sqrt{m^2 n})^2 + (\sqrt{m n^2})^2 = 1$$

$$\left(\sqrt{\left(\frac{\cos^2 \theta}{\sin \theta} \right)^2 \times \frac{\sin^2 \theta}{\cos \theta}} \right)^2 + \left(\sqrt{\frac{\cos^2 \theta}{\sin \theta} \times \left(\frac{\sin^2 \theta}{\cos \theta} \right)^2} \right)^2$$

$$= \left(\sqrt{\frac{\cos^4 \theta}{\sin^2 \theta} \times \frac{\sin^2 \theta}{\cos \theta}} \right)^2 + \left(\sqrt{\frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^4 \theta}{\cos^2 \theta}} \right)^2$$

$$= (\cos \theta)^2 + (\sin \theta)^2$$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1 = \text{RHS}$$

Hence, proved

- 27-
i) 10.7
ii) 4.29 miles
iii) 11.43 miles
iv) 3.5 miles

- 28-
i) (4, 13/3)
ii) $2\sqrt{10}$ units
iii) (13/9, 0)
iv) (5, 4)

29- car in same direction

$$\Rightarrow S = x - y$$

$$D = 100 \text{ Km}$$

$$t = 5 \text{ hrs}$$

$$S = D/T$$

$$\Rightarrow x - y = 20 \quad \text{--- (1)}$$

car in opposite direction

$$\Rightarrow S = x + y$$

$$D = 100 \text{ Km}$$

$$T = 1 \text{ hr}$$

$$S = D/T$$

$$\Rightarrow x + y = 100 \quad \text{--- (2)}$$

$$\text{(1) + (2)}$$

$$\Rightarrow x + y = 100$$

$$+ x - y = 20$$

$$\text{(+) (-) (+)}$$

$$2x = 120$$

$$\boxed{x = 60 \text{ Km/h}}$$

Put $x = 60$ in (1)

$$x - y = 20$$

$$60 - y = 20$$

$$\boxed{y = 40 \text{ Km/h}}$$

30- Given - $\angle ABD = \angle CDB = \angle PQB = 90^\circ$

$$AB = x$$

$$CD = y$$

$$PQ = z$$

To prove - $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

Proof - In $\triangle ADB$ & $\triangle PDQ$

$$AB \parallel PQ,$$

$$\angle ABD = \angle PQD \quad (\text{corresponding})$$

$$\angle ADB = \angle PDQ \quad (\text{common})$$

$$\triangle ADB \sim \triangle PDQ \quad (AA)$$

$$\therefore \frac{DQ}{DB} = \frac{PQ}{AB}$$

$$\Rightarrow \frac{DQ}{DB} = \frac{z}{x} \quad \text{--- (1)}$$

Similarly, $\triangle PBQ \sim \triangle DCB$

$$\text{And, } \frac{BQ}{DB} = \frac{z}{y} \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2}$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = \frac{DQ}{DB} + \frac{BQ}{DB}$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = \frac{DQ + BQ}{DB}$$

$$\Rightarrow z \left(\frac{1}{x} + \frac{1}{y} \right) = \frac{DB}{DB} \cdot 1$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

Hence, proved.